

HTML - HyperText Markup Language



Contents

- Understanding HTML
 - Title, Section Headers
 - New Paragraph, and Line Break
 - Images
 - Hyper Links
 - Tables



HyperText Markup Language (HTML)

- HTML is the markup language for creating web pages, could be hand-written or generated by tools such as Front Page or Dreamweaver
- What does HTML looks like?

`<HTML>`

`<HEAD>`

`<TITLE>My first HTML document</TITLE>`

`</HEAD>`

← *Header
Section*

`<BODY>`

`<P>Hello world!`

`</BODY>`

← *Body
Section*

`</HTML>`

Basic Syntax of a HTML Statement

- Element

- It is a label that identifies and structures the different parts of a web page.
- A **non-empty** element consists of an opening tag, the content, and a closing tag.

Opening tag Content Close tag

`<title>` `HTML Basic` `</title>`

- An **empty** element consists of the initial less than sign, the element's name, a space, and the final greater than sign.

Initial less than sign Element name Final greater than sign

`<br >`

Basic Syntax of a HTML Statement

- Attribute and Value
 - They contain information that further describe the purpose and content of the element.
 - An attribute's value is **enclosed in quotation marks** (*especially when the value contains space*).

Attribute Value
└──┬──┘ └──┬──┘
Log on

Attribute Value for src Attribute Value for width
└──┬──┘ └──┬──┘ └──┬──┘ └──┬──┘

Basic HTML Structure

```
1
2    <!DOCTYPE html>
3    <html>
4        <head>
5            <meta charset="utf-8">
6            <meta name="author" content="" >
7            <meta name="description" content="" >
8            <meta name="keywords" content="" >
9            <title>Untitled Document</title>
10        </head>
11        <body>
12
13        </body>
14    </html>
```

Basic HTML Structure

- Line 3 - `<html>` Element
 - It is efficiently the first element of an HTML/XHTML document.
 - The *(old)* XHTML requires the declaration of its namespace, together with the main language in which your page is written.
(e.g. for hyphenation, spell check, left-to-right order...etc)

The diagram shows the `<html>` element with two attributes. A bracket above the `xmlns="http://www.w3.org/1999/xhtml"` attribute is labeled "XHTML Namespace". Another bracket above the `xml:lang="en"` attribute is labeled "Main Language of an Element".

```
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
```

- In HTML5 we simply writes `<html>`. Meanwhile the `lang` property is bound to individual elements to indicate whether the text is in Chinese / English... etc.. (e.g. `<p lang="en">...</p>`)

Specifying the Language *(if needed)*

- While saving with the proper encoding and declaring that encoding is essential, it can also be useful to specify the language in which your page is written.

- A list of language code is defined in ISO 639:

http://sunsite.berkeley.edu/amher/iso_639.html

Language Name	Code
Chinese	zh
English	en
French	fr
German	de
Japanese	ja
Korean	ko

- Instead of binding **lang** to individual element (*preferred*), an alternative way is to put the followings inside **<head>**:

```
<meta http-equiv="content-language" content="en" >
```

Basic HTML Structure

- **<head> Element**

- It provides an unordered collection of information about the document, which could be useful for browsers.

Comment Element	Description
base	It specifies an explicit URL used to resolve links and references to external sources, such as images, and style sheets. (e.g. <code><base href="http://www.cs.cityu.edu.hk/" /></code>)
link	It enables the current document to establish links to external documents, such as style sheets.
meta	It conveys hidden information about the document to server & client.
script	It specifies a script for the page that is interpreted by a script engine.
style	It specifies a style sheet for the page.
title	It contains the title of the document.

Specifying Author, Description and Keywords

- The meta description tag allows you to influence the description of your HTML page in the crawlers that support the tag.
- The meta keywords tag allow you to provide additional text for crawler-based [search engines](#) to index along with the body copy.
- Hence, the quality of search results may be improved.

```
<meta name="author" content="..." />  
<meta name="description" content="..." />  
<meta name="keywords" content="..." />
```

Specifying Author, Description and Keywords

```
3    <html>
4        <head>
5            <meta charset="utf-8">
6            <meta name="author" content="ebanking team" >
7            <meta name="description" content="Our online payment services
8                provide a simple and easy way to manage your bills" >
9            <meta name="keywords" content="payment, bill payment, autopay" >
10           <title>Untitled Document</title>
11       </head>
12       <body>
13       </body>
14   </html>
```

Declaring the Encoding

- The most commonly used character encoding is **Unicode**.
 - It is a universal system for encoding all of the characters in all of the world's languages.
 - Each character in each language is assigned a unique code.
 - Unicode Code Charts <http://www.unicode.org/charts/>

A	À	Á	Â	一	城	づ	う
1D00	00C0	00C1	00C2	4E00	57CE	3065	3046

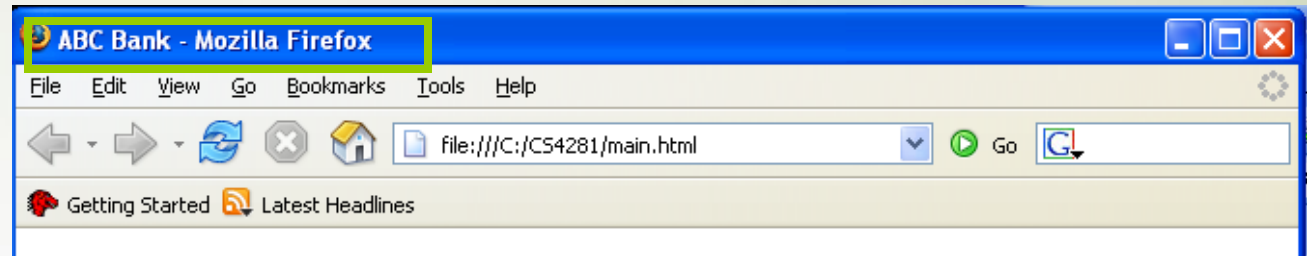
- UTF-8 is preferred over regional encoding (e,g, BIG5 / GB2312) as it supports characters from different languages *at the same time*:

`<meta charset="utf-8">`

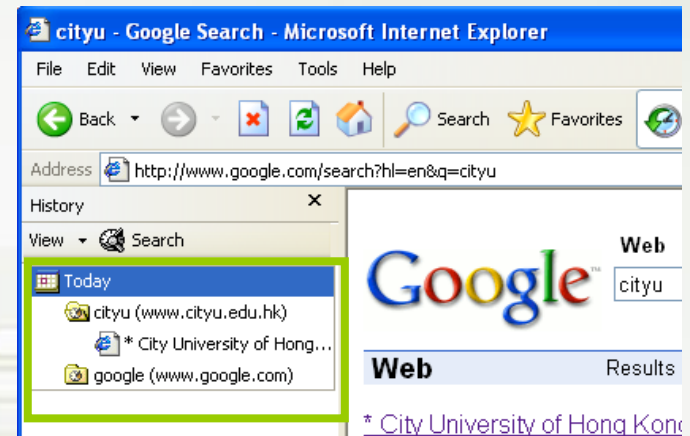
- The encoding you declare must match the encoding with which your page was saved. Otherwise, characters that differ between the encodings will display incorrectly.

Creating a Title

- In most browsers, the title appears in the title bar of the window.



- A title should be short and descriptive.
 - It affects page ranking in search engines.
 - It is used in History lists, Favorites lists, and Bookmarks menus to identify the page.
- Depending on the `<!DOCTYPE>` chosen, `<title>` could be **COMPULSORY**. (e.g. in HTML5)
- `<title>` can only exist inside `<head>`, not `<body>`



`<title> ... </title>`

Creating a Title

```
3  <html >
4      <head>
5          <meta charset="utf-8">
6          <meta name="author" content="ebanking team" >
7          <meta name="description" content=" Our online payment services
provide a simple and easy way to manage your bills " >
8          <meta name="keywords" content="payment, bill payment, autopay" >
9              <title>ABC Banking Corporation Limited</title>
10         </head>
11         <body>
12
13         </body>
14     </html>
```

Creating Section Headings / sub-headings

- It briefly describes the topic of the text section it introduces.
- Heading information may be used by user agents, for example, to construct a **table of contents** for a document automatically.
- It provides for up to six levels of **headings / sub-headings** in the Web page.

<hn> ... </hn>

where **n** is a number from 1 to 6

- **<h1>** is greater than **<h2>**
<h2> is greater than **<h3>**...
however the exact size and style may varies slightly from browser to browser

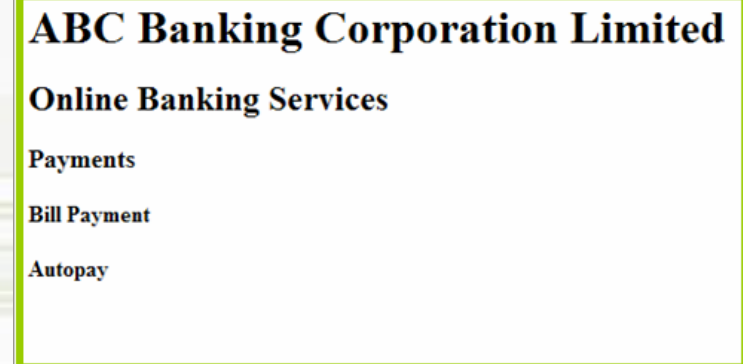
<h1> ... </h1>



Creating Section Headers

```
4 <head>
5   <meta charset="utf-8">
6   <meta name="author" content="ebanking team" >
7   <meta name="description" content=" Our online p..." >
8   <meta name="keywords" content="payment, bill payment, autopay" >
9   <title>ABC Banking Corporation Limited</title>
10 </head>
11 <body>
12   <h1>ABC Banking Corporation Limited</h1>
13   <h2>Online Banking Services</h2>
14   <h3>Payments</h3>
15   <h4>Bill Payments</h4>
16   <h5>AutoPay</h5>
17 </body>
```

Larger font <h1>



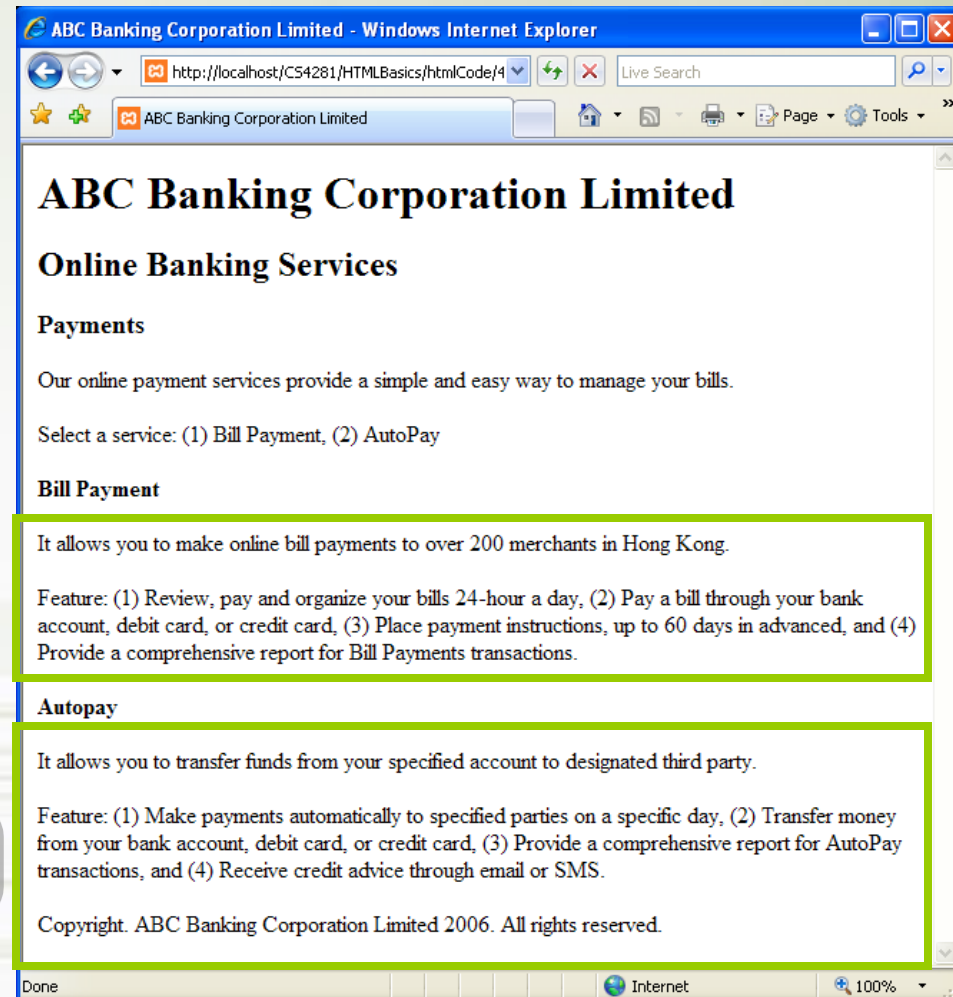
ABC Banking Corporation Limited
Online Banking Services
Payments
Bill Payment
Autopay

Smaller font <h5>

Starting a New Paragraph

- HTML **does not recognize** the **<Enter>** or other extra **white space** that you enter in your text editor.
- The opening and closing **p** tags are used to enclose each paragraph.

`<p> ... </p>`



Starting a New Paragraph

11 <body>

12 <h1>ABC Banking Corporation Limited</h1>

13 <h2>Online Banking Services</h2>

14 <h3>Payments</h3>

15 <p>What are payments? </p>

16 <p>Our online payment services provide a simple and easy way to</p>

17 <p>Select an account: (1) Bill Payment, (2) AutoPay </p>

18
19 <h4>Bill Payments</h4>

20 <p>It allows you to make online bill payments to over 200 merchants in ... </p>

21 <p>Feature: (1) Review, pay and organize your bills 24-hour a day, (2) Pay a bill through your bank account, debit card, or credit card, (3) Place payment instructions, up to 60 days in advanced, and (4) Provide a comprehensive report for Bill Payments transactions. </p>

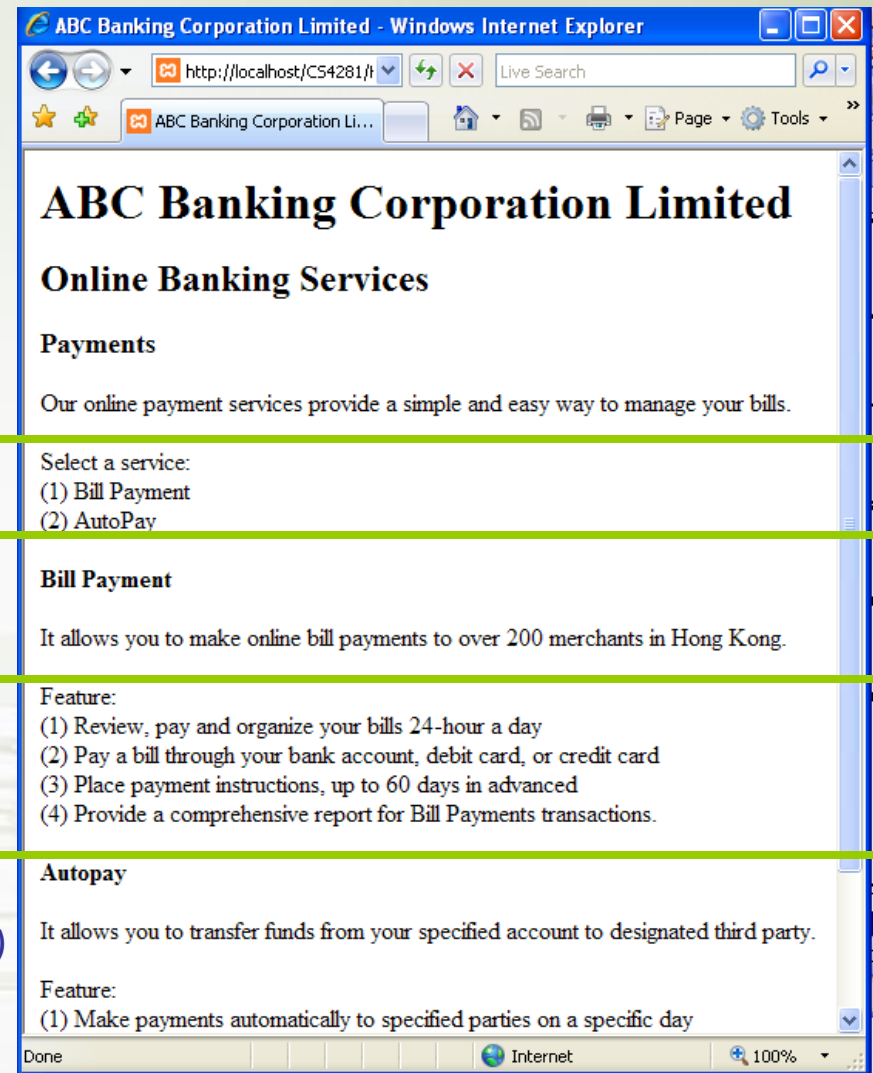
22
23 <h4>AutoPay</h4>

Creating a Line Break

- Browsers automatically wrap text according to the width of the block or window, creating new lines as necessary.
- The `br` tag can be used to create **manual** line breaks anywhere you like.
- Lines of text appear one after another.

`<br />`

The slash `/` in `<br/>` (and other empty tags) is optional in HTML, but is a must for (old) XHTML documents



Creating a Line Break

11 <body>

12 <h1>ABC Banking Corporation Limited</h1>

13 <h2>Online Banking Services</h2>

14 <h3>Payments</h3>

15 <p>What are payments? </p>

16 <p>Our online payment services provide a simple and easy way to manage your bills.</p>

17 <p>Select an account: **
** (1) Bill Payment **
** (2) AutoPay </p>

18
19 <h4>Bill Payments</h4>

20 <p>It allows you to make online bill payments to over 200 merchants in Hong Kong.</p>

21 <p>Feature: **
** (1) Review, pay and organize your bills 24-hour a day **
** (2) Pay a bill through your bank account, debit card, or credit card **
** (3) Place payment instructions, up to 60 days in advanced **
** (4) Provide a comprehensive report for Bill Payments transactions. </p>

Adding Comments

- Text inside `<!--` and `-->`
- `<!--` and `-->` could be placed on different lines (*for long comments*)
- Not visible unless user choose to view source
- Ignored by browser and hence not affecting page appearance
- Only for developers' reference

Comment is only for human reference and is ignored by the browser



```
view-source: - Source of: http://localhost/CS4281/HTM
File Edit View Help

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1
<html xmlns="http://www.w3.org/1999/xhtml" >
<head>
<meta http-equiv="Content-Type" content="tex
<meta name="author" content="ebanking team"
<meta name="description" content="Our online
<meta name="keywords" content="payment, bill
<title>ABC Banking Corporation Limited</titl
</head>
<body>
<!-- Page Header -->
    <h1>ABC Banking Corporation Limited<
    <h2>Online Banking Services</h2>
<!-- Page Content -->
    <h3>Payments</h3>
```

Images

Inserting Images on a Page

- Common image formats can be placed on the Web page:
 - Windows Bitmap (BMP)
 - Windows Enhanced Metafile (EMF)
 - Graphic Interchange Format (GIF)
 - Joint Photographic Experts Group (JPG/JPEG)
 - Portable Network Graphics (PNG)
 - Scalable Vector Graphic (SVG)
(for line drawings like charts)

```

```



Images

Inserting Images on a Page

```
11 <body>
12 <!-- Page Header -->
13     <h1>ABC Banking Corporation Limited</h1>
14     <h2>Online Banking Services</h2>
15
16 <!-- Related Products Information -->
17     <p>Information</p>
18     <p><br />Loans</p>
19
20 <!-- Page Content -->
21     <h3>Payments</h3>
22     <p>Our online payment services provide a simple and easy way to manage your
23     bills.</p>
24     <p>Select a service: <br /> (1) Bill Payment <br /> (2) AutoPay </p>
```

Images

Absolute URLs vs. Relative URLs

- Uniform Resource Locator (URL)
 - It simply means “web address”.
 - It contains information about where a file is and what a browser should do with it.

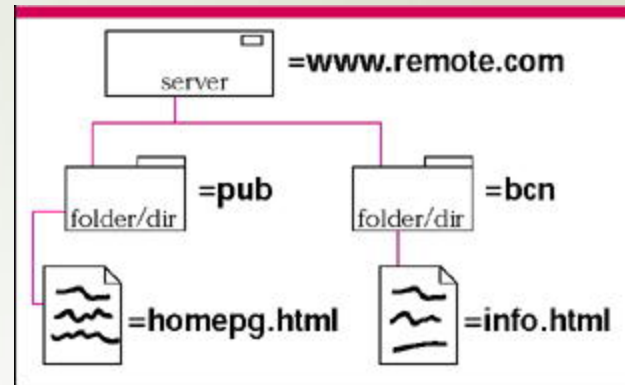
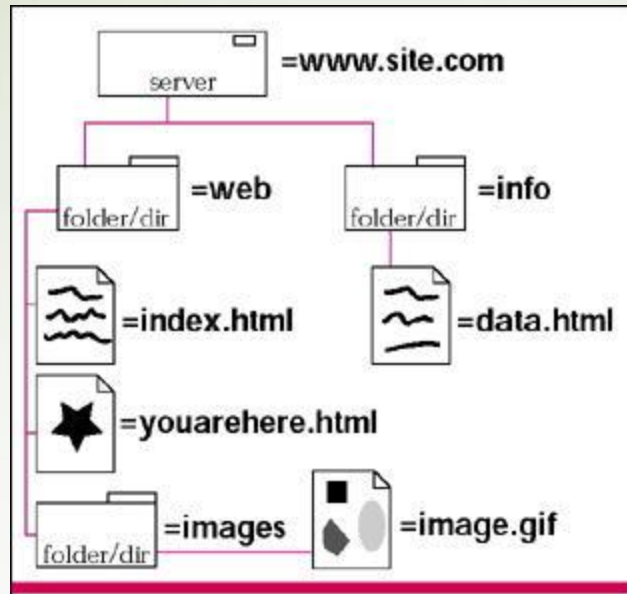
Protocol **Server name** **Path** **File name**

`http://www.cityu.edu.hk/cityu/images/top_left.gif`

- Absolute URL
 - It shows the **entire path** to the file, including the scheme, server name, the complete path, and the file name itself.
 - It is generally used for
 - Referencing a file from someone else’s server
 - FTP sites, newsgroups, and email address
- Relative URL
 - It describes the location of the desired file with **reference to the location of the file** that contains the URL itself.
 - **.** (*single dot*) means “current folder” and **..** (*double dot*) means “parent folder”
 - It is generally used for the files which is
 - In the same/neighboring directory / subdirectory on the same server

Images

Absolute URLs vs. Relative URLs



File name	Absolute URL (can be used anywhere)	Relative URL (only works in yourarehere.html)
index.html	www.site.com/web/index.html	index.html
image.gif	www.site.com/web/images/image.gif	images/image.gif
data.html	www.site.com/info/data.html	../info/data.html
homepg.html	www.remote.com/pub/homepg.html	(none: use absolute)
info.html	www.remote.com/bcn/info.html	(none: use absolute)

Images

Scaling an Image *(legacy method)*

- The size of an image can be changed by specifying new height and width values for the image.
 - Pixels (color dot)
 - Percentages
- Percentages are of the available width or height that the image could fill, usually the width or height of the current window.
- If the width property of an image is specified, but not the height property, the resulting height of the image is sized **proportionally** to the specified width.

```

```

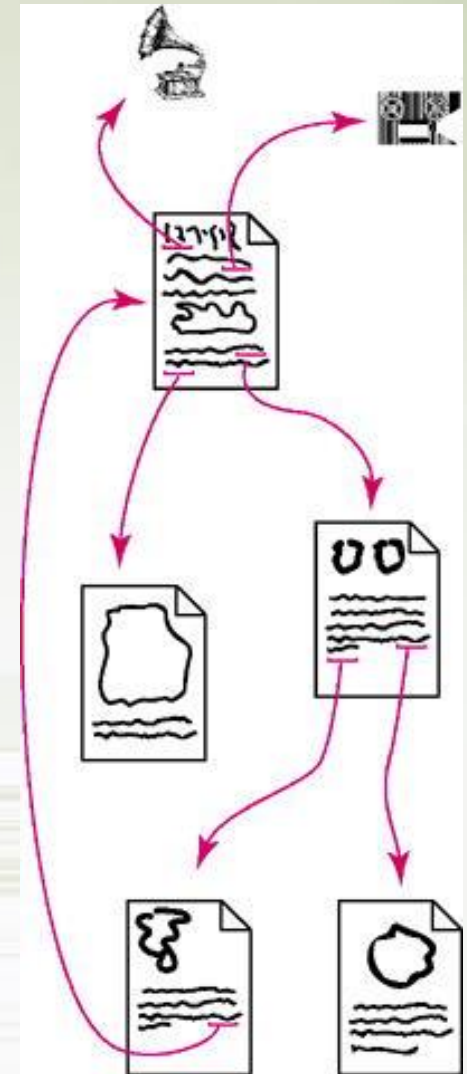
Images

Scaling an Image

```
11 <body>
12 <!-- Page Header -->
13     <h1>ABC Banking Corporation Limited</h1>
14     <h2>Online Banking Services</h2>
15
16 <!-- Related Products Information -->
17     <p>Information</p>
18     <p>
        <br />Loans</p>
19
20 <!-- Page Content -->
21     <h3>Payments</h3>
22     <p>Our online payment services provide a simple and easy way to manage your
bills.</p>
23     <p>Select a service: <br /> (1) Bill Payment <br /> (2) AutoPay </p>
```

Links

- Links are the distinguishing features of the WWW.
- You will be able to
 - Skip from one session / page to another
 - Download files with HTTP / FTP
- A link has three parts:-
 - **Href:** It specifies the **resource** to be accessed when the visitor clicks the link, e.g. show an image, play a sound, send an email...
 - **Label:** It is the part the visitor **sees and clicks on** to reach the destination. It can be text, an image or both.
 - **Target:** It determines **where** the destination will be displayed. It might be a particular named window or frame.

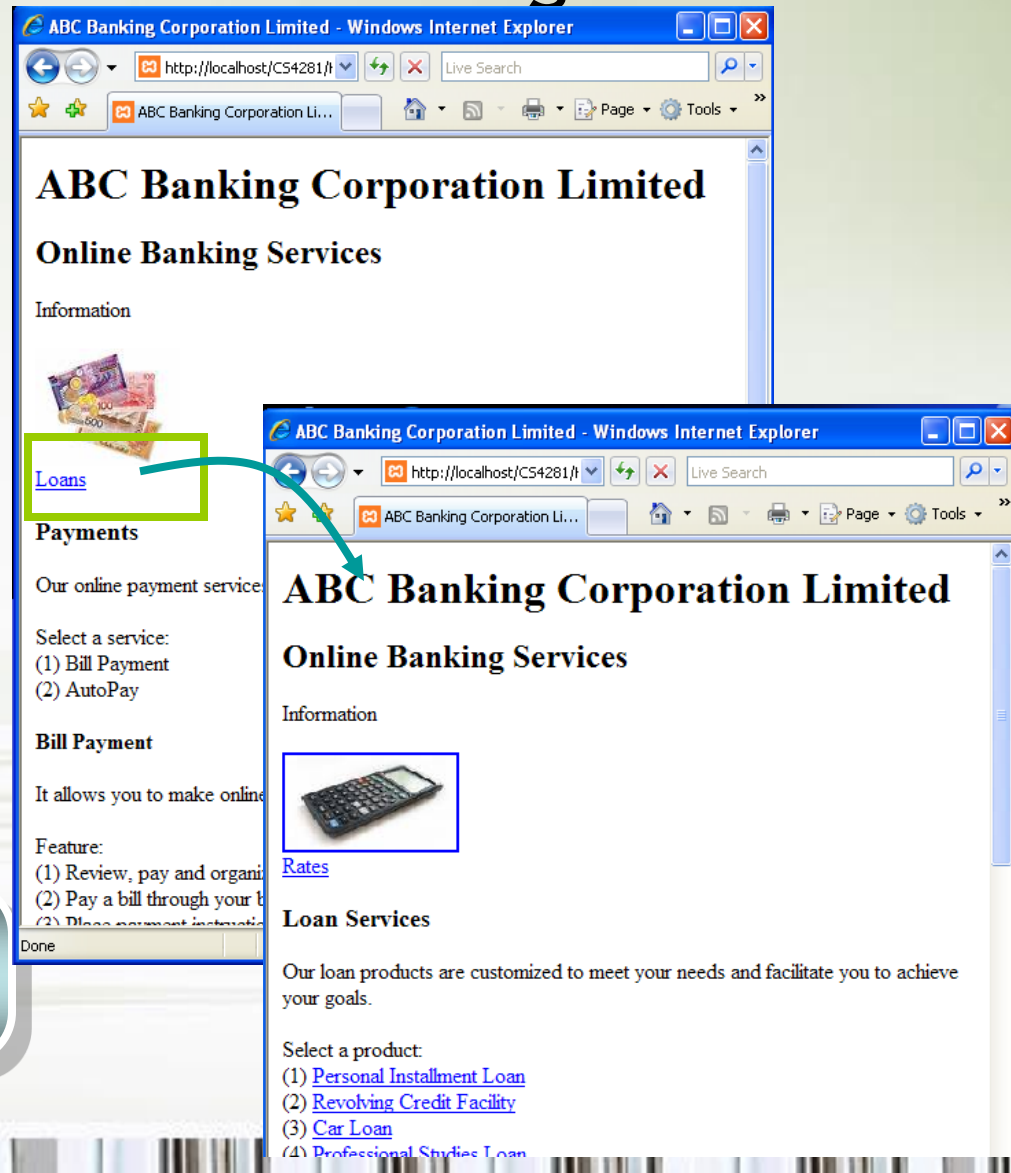


Links

Creating a Link to Another Web Page

- You can create links from one page to the next.
- Recommendations
 - Note that the address are **case sensitive** (i.e. Uppercase is different from lowercase).
 - Use the keyword(s) as the link's label.
 - Don't make the link's label too long.

```
<a href="destination">Label </a>
```



Links

Creating a Link to Another Web Page

```
11 <body>
12 <!-- Page Header -->
13 <h1>ABC Banking Corporation Limited</h1>
14 <h2>Online Banking Services</h2>
15
16 <!-- Related Products Information -->
17 <p>Information</p>
18 <p><br /><a href="loans.html">Loans</a></p>
19
20 <!-- Page Content -->
21 <h3>Payments</h3>
22 <p>Our online payment services provide a simple and easy way to manage your
bills.</p>
23 <p>Select a service: <br /> (1) Bill Payment <br /> (2) AutoPay </p>
```

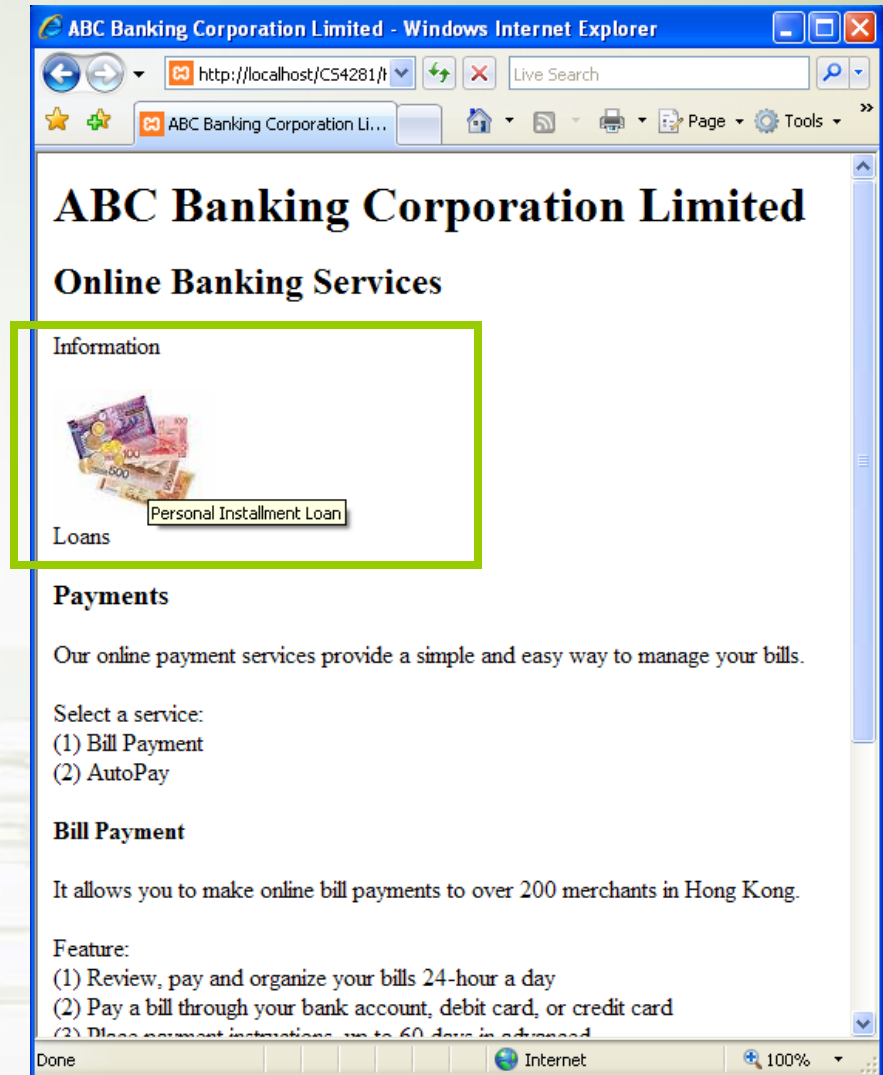
Images

Offering Alternate Text

- Descriptive text can be added if the image does not appear.
(old browser like Lynx, or in case image missing / damaged)
- Although browsers could still show the image without *alt*, but in theory the *alt* attribute is compulsory.

```

```



Images

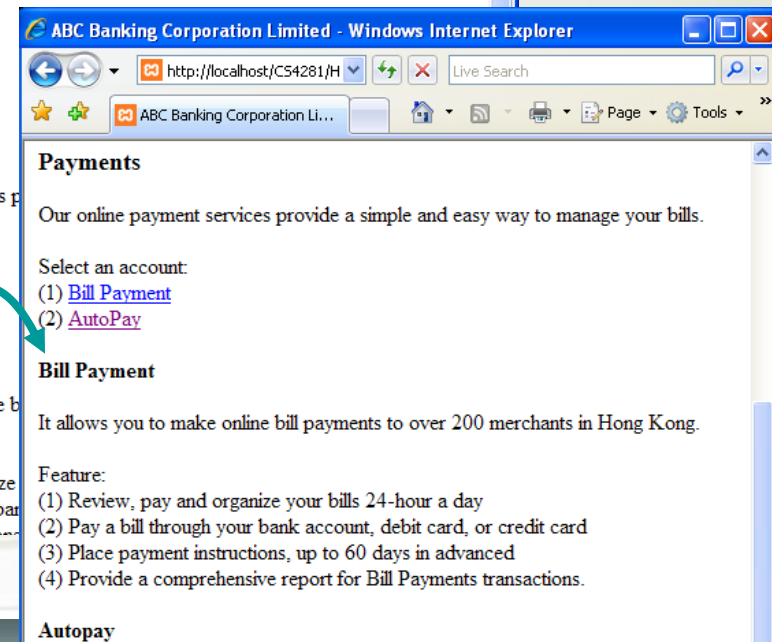
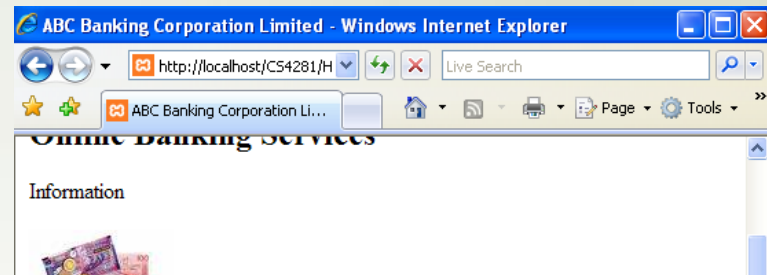
Offering Alternate Text

```
11 <body>
12 <!-- Page Header -->
13     <h1>ABC Banking Corporation Limited</h1>
14     <h2>Online Banking Services</h2>
15
16 <!-- Related Products Information -->
17     <p>Information</p>
18     <p><br
19 />Loans</p>
20 <!-- Page Content -->
21     <h3>Payments</h3>
22     <p>Our online payment services provide a simple and easy way to manage your
23 bills.</p>
24     <p>Select a service: <br /> (1) Bill Payment <br /> (2) AutoPay </p>
```

Links

Creating a Link to Specific Locations on Web Page

- An anchor is the text you will use internally to identify that section of the Web page.
- Once you have created an anchor, you can define a link so that a visitor's click will bring them directly to the section of the Web page, that contains the anchor, not just the top of that page.



`id="anchorname" or <a name="anchorname">...</a>`
`<a href="page.html#anchorname">Label</a>`

Links

Creating a Link to Specific Locations on Web Page

```
20 <!-- Page Content -->
```

```
21 <h3>Payments</h3>
```

```
22 <p>Our online payment services provide a simple and easy way to manage your  
bills.</p>
```

```
23 <p>Select an account: <br />
```

```
24 (1) <a href="#billPayment">Bill Payment</a> <br />
```

```
25 (2) <a href="#autopay">AutoPay</a> </p>
```

```
26  
27 <h4 id="billPayment">Bill Payment</h4>
```

```
28 <p>It allows you to make online bill payments to over 200 merchants in Hong Kong.  
</p>
```

```
29 <p>Feature: <br/> (1) Review, pay and organize your bills 24-hour a day <br /> (2)  
Pay a bill through your bank account, debit card, or credit card <br /> (3) Place payment  
instructions, up to 60 days in advanced <br /> (4) Provide a comprehensive report for Bill  
Payments transactions. </p>
```

```
30  
31 <h4 id="autopay">Autopay</h4>
```

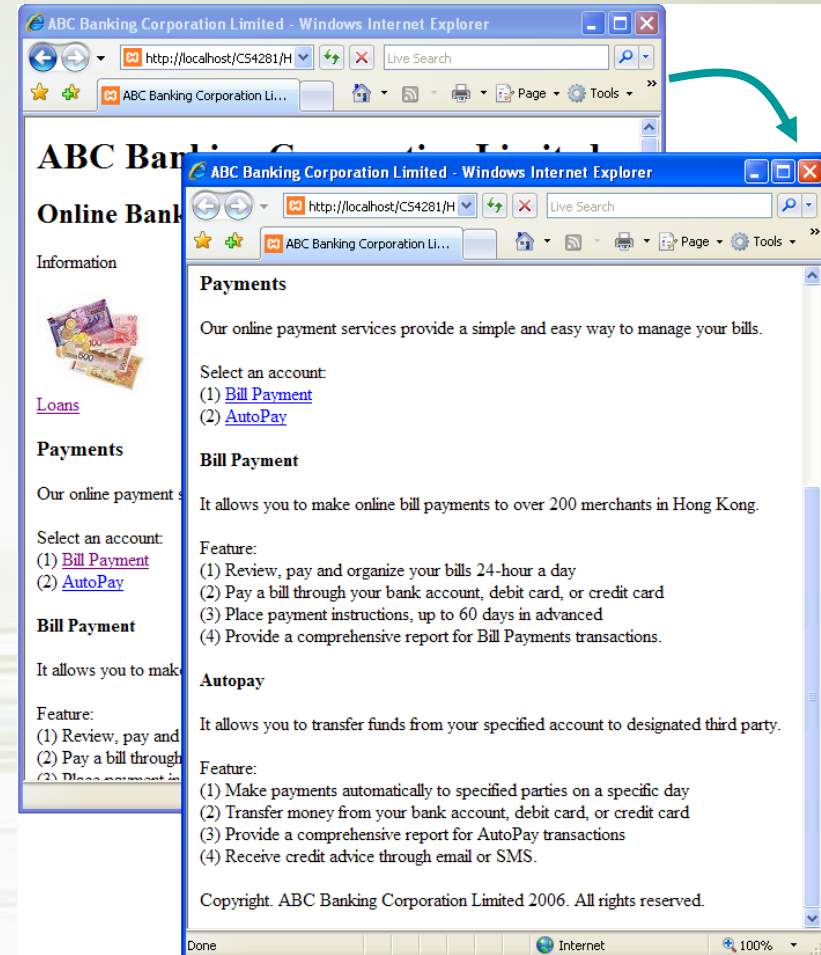
Links

Targeting Links to Specific Windows

- A target attribute is used to open a link in a particular frame or even in a new window.
- NOTE: Instead of using the **target** attribute, HTML documents may also use JavaScript as an alternative (*e.g. for opening new advertisement popup window*) . It will be discussed later.

`target="window_name"` or
`target="_blank"`

New window is prompted



Links

Targeting Links to Specific Windows

1
2 `<!DOCTYPE html>`

3 `<html>`

4 `<head>`

...

22 `<p>Our online payment services provide a simple and easy way to manage your bills.</p>`

23 `<p>Select an account: <br />` The page will be opened in a new window.

24 (1) `<a href="#billPayment" target="_blank">Bill Payment</a> <br />`

25 (2) `<a href="#autopay">AutoPay</a> </p>`

26
27 `<h4 id="billPayment">Bill Payment</h4>`

28 `<p>It allows you to make online bill payments to over 200 merchants in Hong Kong. </p>`

Links

Creating Other Kinds of Links

- Movies / Sounds / Programs / Excel Spreadsheets, etc.

```
<a href="http://www.site.com/path/file.ext">Label</a>
```

- FTP Site

```
<a href="ftp://ftp.site.com/path">Label</a>
```

- Newsgroup

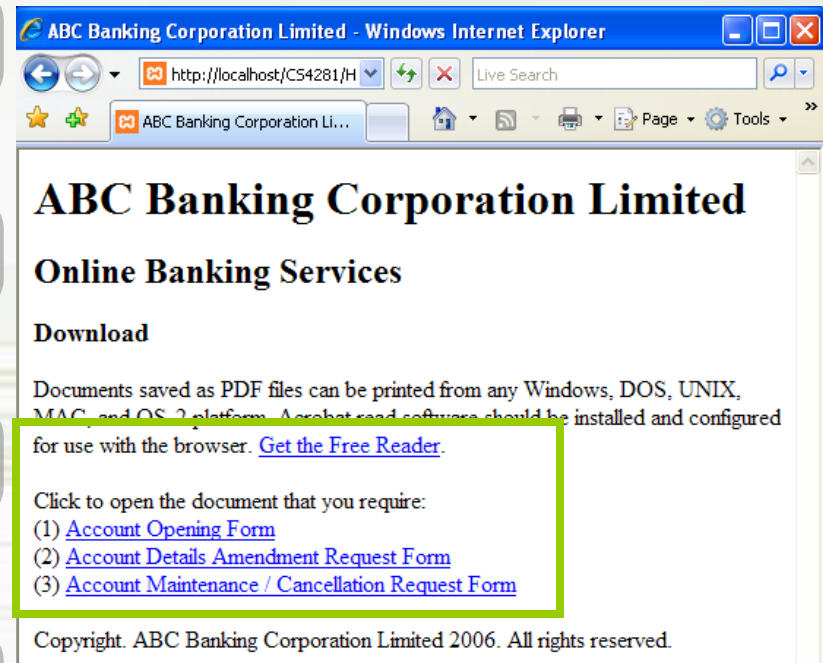
```
<a href="news:newsgroup">Label</a>
```

- Email

```
<a href="mailto:name@site.com">Label</a>
```

- Telnet Site

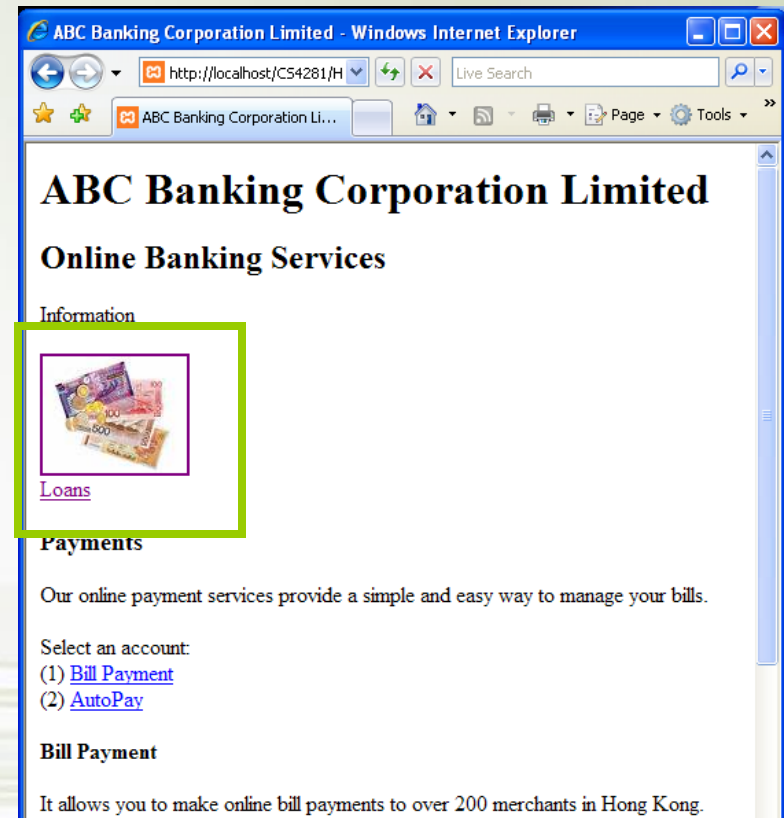
```
<a href="telnet://site">Label</a>
```



Links

Using Images to Label Links

- Adding an image to a link creates a navigational button that the visitor can click to access the referenced URL.
- Simply put the `<img>` tag in between `<a>` and `</a>`
- IE automatically draw a border around image links but most browsers do not.



```
<a href="destination.html"></a>
```

Tables

- Tables are the best way for displaying tabular information.
- Major Components
 - Table Cells
 - They are basic structural unit of tables.
 - They contain data.
 - Table Rows
 - They are horizontal set of cells.
 - They are grouped into a head, body, and foot sections.
 - Table Columns
 - They are vertical set of cells.

Diagram illustrating the structure of a table with 4 rows and 8 columns. The table is represented by a grid of cells, each containing the letter 'C'. Arrows indicate the rows and columns.

Column

1 2 3 4 5 6 7 8

Row 1 →

Row 2 →

Row 3 →

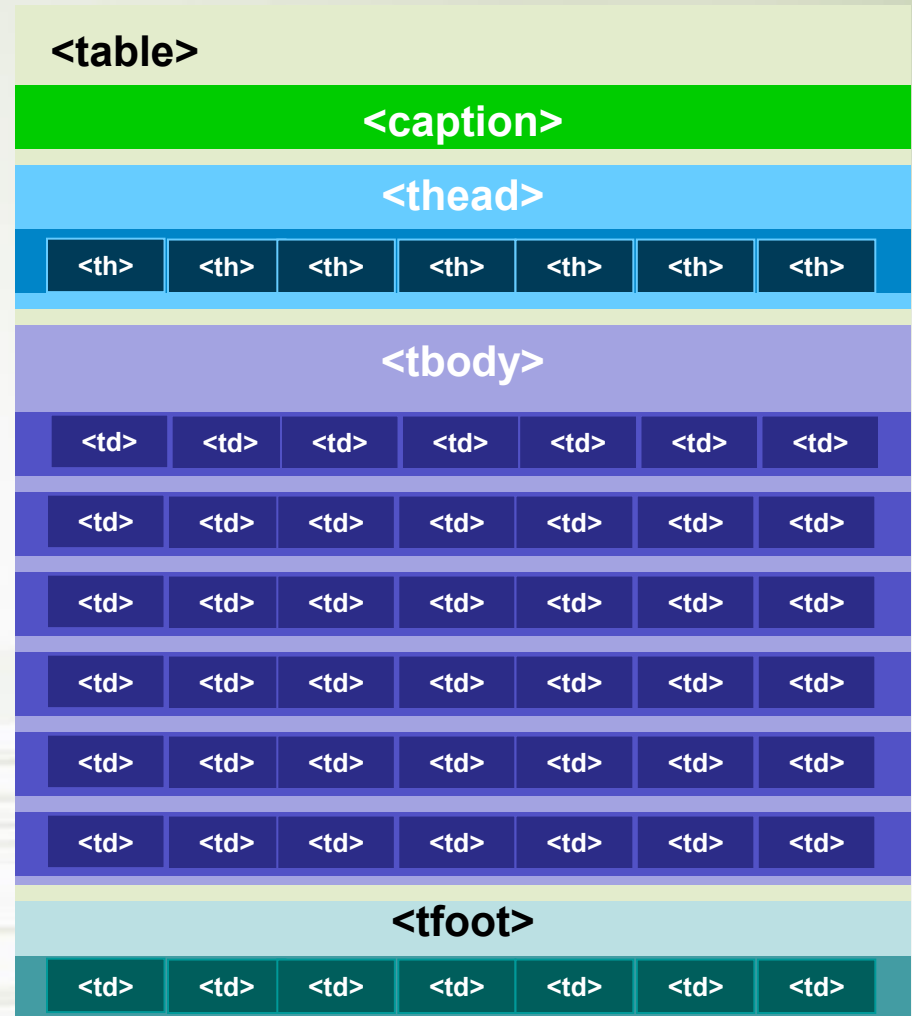
Row 4 →

Cell C

C	C	C	C	C	C	C	C
C	C	C	C	C	C	C	C
C	C	C	C	C	C	C	C
C	C	C	C	C	C	C	C

Simple Table Structure

```
<table summary=" ... ">
  <caption> ... </caption>
  <thead>
    <tr><th> ... </th> </tr>
  </thead>
  <tfoot>
    <tr><td> ... </td></tr>
  </tfoot>
  <tbody>
    <tr><td> ... </td></tr>
    <tr><td> ... </td></tr>
    <tr><td> ... </td></tr>
    ...
  </tbody>
</table>
```



Simple Table Structure

- `<table>` Element
 - It contains all other elements that specify caption, rows, content and formatting.
- summary Attribute
 - It is a summary of a table's content and structure so that people using non-visual user agents may better understand it.
- `<caption>` Element
 - It provides a short description of the table's purpose.
 - It is only permitted immediately after the `<table>` start tag.
 - A `<table>` element may only contain at most one `<caption>` element.
- `<thead>` Element
 - It contains table head information (*e.g. col label*) about the table's columns.
- `<tfoot>` Element
 - It contains table foot information (*e.g. total*) about the table's columns.

Simple Table Structure

- `<tbody>` Element
 - It contains the main information for the table (*excluding headings*).
- `<tr>` Element
 - It acts as a container for a row of tables cells.
- `<th>` Element
 - It defines a cell that contains header information text are **bolded and centered**.
- `<td>` Element
 - It defines a cell that contains typical data. text are normal: left-aligned and not bolded .

How to merge TABLE cells ?

- The `<th>` and `<td>` elements have `colspan` and `rowspan` attributes which allow merging of columns and/or rows.

```
<table border=1>
  <tr> <td>1</td> <td>2</td> <td>3</td> </tr>
  <tr> <td>4</td> <td colspan="2">5 and 6</td> </tr>
  <tr> <td>7</td> <td>8</td> <td>9</td> </tr>
</table>
```

1	2	3
4	5 and 6	
7	8	9

```
<table border=1>
  <tr> <td colspan="2" rowspan="2">*</td> <td>3</td> </tr>
  <tr>                                     <td>6</td> </tr>
  <tr> <td>7</td>          <td>8</td>          <td>9</td> </tr>
</table>
```

*		3
		6
7	8	9

Other attributes of <table> and <td>

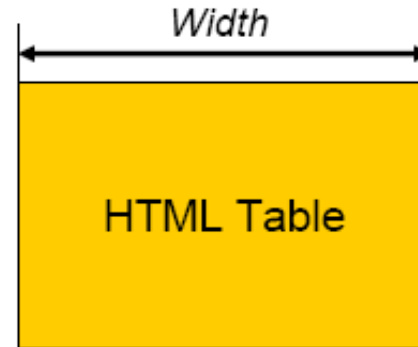
- Apart from **colspan** and **rowspan**, there are many other attributes for the **<table>** / **<td>** / **<th>** tags, which can be used to customize the table appearance.
- Yet those attributes of the **<table>** / **<td>** tags are becoming **deprecated**. Students are advised to switch to CSS statements for new pages instead.
(Engineers believe that formatting should be done in CSS, but not HTML, which is supposed to be used for structure)
- Interested students could refer to:
<http://www.somacn.com/p141.php>

Attributes of <table> Tag



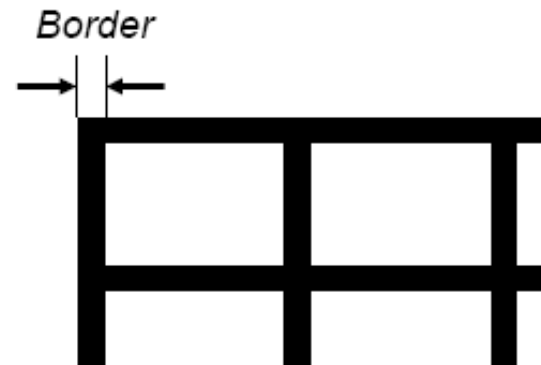
● WIDTH

- Defines the width of the table



● BORDER

- Defines the size of the border

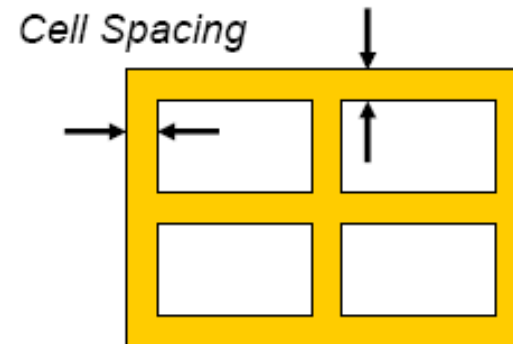


Attributes of <table> Tag



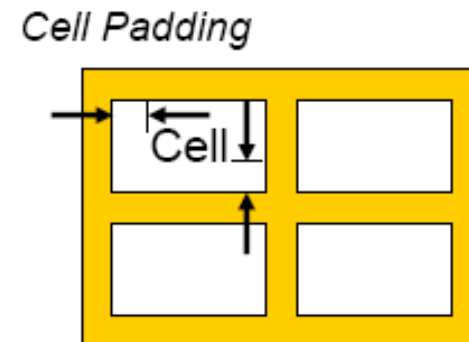
- CELLSPACING

- Defines the space separating two table cells



- CELLPADDING

- Defines the space around the content in a table cell



Attributes of <td> Tag



- WIDTH, HEIGHT

- Define the width and height of a cell in number of pixels

- COLSPAN

- Defines how many columns a cell occupies

- ROWSPAN

- Defines how many rows a cell occupies

Column Span is 2

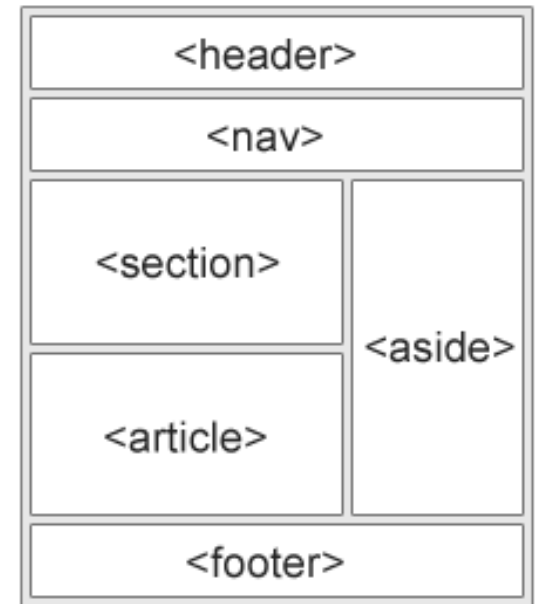
Row Span is 2

Attributes of <td> Tag

- ALIGN
 - Same as the TR element, defines the alignment of content
- VALIGN
 - Same as the TR element, defines the vertical alignment of content
- NOWRAP
 - Indicates no line wrap is allowed in a cell

HTML5 semantic tags

- HTML is designed to **describe the content** (the **WHATs**) of a webpage, it may happen that a tag **does not make any visible change** to the appearance of webpage.
- HTML5 introduced some new semantic tags to **describe the meaning** of the parts of your webpage:
- Those tags are not compulsory, and the nesting order is not specified as well (*e.g. <section> inside <article>, or <article> inside <section> are both okay*)



HTML5 semantic tags

```
<body>
  <nav>
    <a href="#">Link 1</a>
    <a href="#">Link 2</a>
    <a href="#">Link 3</a>
  </nav>
  <article>
    <h1>Introduction</h1>
    <p>In this document, we're going to discuss...</p>
  </article>
</body>
```

`<nav>` is saying that this block contains links

`<article>` is saying that this block contains standalone content

By default, the appearance is the same with or without the `<nav>` and `<article>` semantic tags

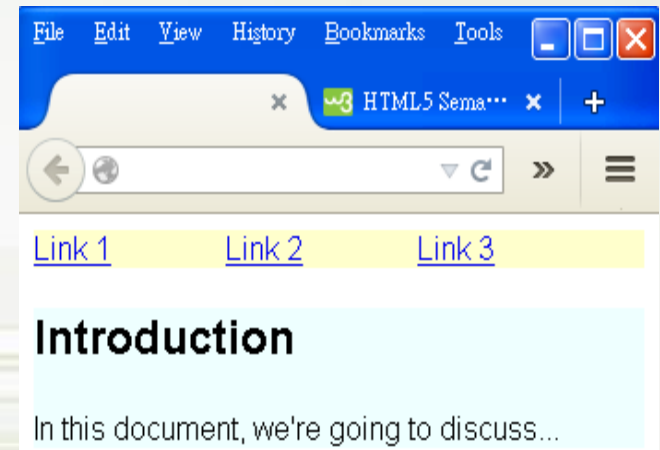
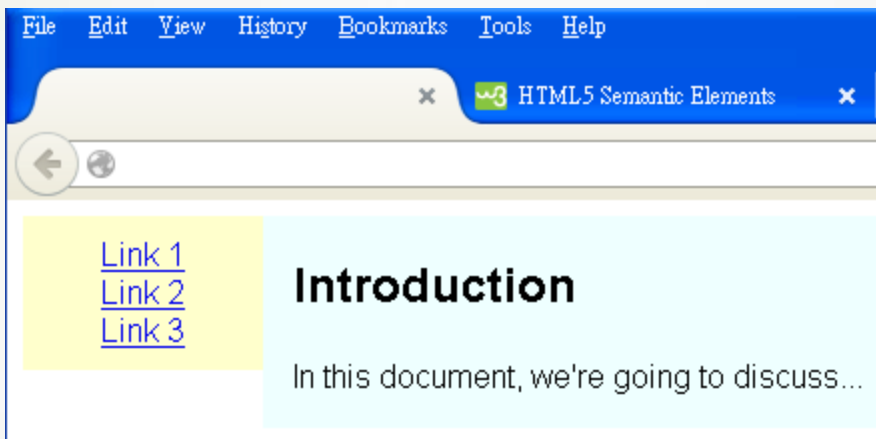
[Link 1](#) [Link 2](#) [Link 3](#)

Introduction

In this document, we're going to discuss...

HTML5 semantic tags

- However, when CSS is applied, programmers can fine-tune the appearance in different scenarios. *(will be taught in later weeks)*
- *For example: put the <nav> links on left if the screen is wide enough, otherwise, put it on the top for narrow screens in mobile...etc*



References

- Jon Duckett, “Beginning HTML, XHTML, CSS, and JavaScript”, Chapter 1 to 4 (p1-21, 45-47, 55-80, 131-158).
- Castro, E. “*HTML for the World Wide Web with XHTML and CSS: Visual QuickStart Guide*”. Berkeley: Peachpit Press.
- W3C HTML Working Group. HTML 5.0. Available from <http://www.w3.org/TR/html5/>
- W3Schools HTML Tutorial. Available from <http://www.w3schools.com/html/>
- Free Online HTML Editor <https://html-online.com/>